

Data-Driven Decision Making: Business Intelligence in Construction Management Efficiency

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Abstract

Construction SMEs face ongoing challenges in cost control, schedule adherence, and resource efficiency despite generating large volumes of data. This work hypothesis that integrating Business Intelligence (BI) into an existing Enterprise Resource Planning (ERP) system improves management efficiency by transforming fragmented data into actionable information. The relevance of this approach is justified by the sector's high levels of material waste and decision-making inefficiencies, often caused by limited analytical support. The main goal of this work is to design, implement, and validate a BI module tailored to the specific needs of construction SMEs, complementing a pre-existing ERP platform. The solution was developed using a C#/.NET backend with Entity Framework Core and MySQL, and a React/TypeScript frontend with Ant Design, following an agile methodology. The final system provides interactive dashboards focused on budget execution, global performance indicators, and project-specific analyses, enabling early detection of deviations in cost and time. The results indicate that improved analytical support and operational transparency provide a foundation for data-driven strategy, advancing the digital maturity and competitive positioning of construction SMEs.

Keywords: Business Intelligence, Enterprise Resource Planning, Construction Industry, Decision Support Systems

1 Introduction

The construction industry faces persistent challenges related to cost control, schedule compliance, and efficient resource allocation, particularly in small and medium-sized enterprises (SMEs). In Portugal, as in many European markets, construction companies operate in environments characterized by complex projects, tight margins, and stringent regulatory requirements. Within this context, the availability of timely and reliable information is critical for effective management. Business Intelligence (BI) has therefore emerged as a strategic approach for transforming operational data into actionable insights, enabling organizations to strengthen decision-making processes through data-driven practices.

The main objective of this work is to design, implement, and validate a Business Intelligence module integrated into an existing Enterprise Resource Planning (ERP) system, specifically tailored to the operational and managerial needs of construction SMEs. Rather than proposing a standalone analytical solution, the developed module extends the analytical capabilities of a pre-existing ERP platform used by a Portuguese construction company. By leveraging consolidated operational data already managed within the ERP, the proposed approach enables the application of Business Analytics techniques, such as executed budgets, to monitor performance, identify inefficiencies, and support both operational control and strategic planning.

To achieve this objective, the work focuses on several specific goals: (i) defining a set of relevant, data-driven performance indicators aligned with construction project management practices; (ii) implementing interactive dashboards and analytical reports to support budget execution monitoring, global performance assessment, and project-specific analysis; and (iii) ensuring seamless integration with existing workflows to minimize disruption and adoption barriers. By complementing an established resource planning system, the proposed BI module contributes to improved transparency, early detection of cost and schedule deviations, and enhanced decision support.

The remainder of this article is structured as follows: Section 2 presents the state of the art, including a comparison with software solutions in the construction sector; Section 3 describes the user stories, functional requirements, adopted technologies, and module development; Section 4 discusses the obtained results; and Section 5 presents the conclusions and outlines future work.

2 State of the Art

This section addresses four areas: data analysis paradigms and Business Intelligence, BI architectures and analytical components, data-driven practices in the construction industry, and BI solutions for construction projects. Together, these perspectives contextualize the research problem and inform the design of the proposed solution.

2.1 Data Analysis Paradigms and Business Intelligence

Data analysis in business contexts is commonly structured into five paradigms: Descriptive, Diagnostic, Predictive, Prescriptive, and Cognitive. These approaches

range from summarizing historical data to explaining causes, forecasting future scenarios, recommending actions, and uncovering complex patterns through advanced artificial intelligence techniques [1]. Their combined application strengthens organizations' ability to anticipate trends, reduce uncertainty, and support evidence-based decision-making [2]. Within this framework, Business Intelligence and Analytics have emerged as the dominant response to Big Data challenges, particularly those related to volume, velocity, and variety [3]. BI encompasses integrated practices and technologies for collecting, processing, analyzing, and disseminating data, enabling the transformation of raw data into actionable knowledge that supports strategic management [4–6].

2.2 Business Intelligence Architectures and Analytical Components

Business Intelligence architectures typically rely on extract, transform and load processes to integrate data from heterogeneous sources [4]. These processes feed Data Warehouses, which act as centralized repositories optimized for historical analysis and complex queries [7]. On top of this infrastructure, analytical techniques such as data mining and online analytical processing enable pattern discovery, predictive modeling, and multidimensional data exploration [1, 8]. The results are commonly presented through dashboards and reports, which consolidate key indicators and support managerial decision-making [9].

2.3 Data-Driven Practices in the Construction Industry

The construction industry generates large volumes of heterogeneous data across the project lifecycle, originating from organizational systems, monitoring platforms, simulations, and unstructured sources [9, 10]. Despite this potential, the sector exhibits relatively low levels of digital maturity, influenced by organizational resistance, cultural factors, and a shortage of qualified professionals [11, 12]. Consequently, BI adoption has often been limited to isolated applications, such as cost control, schedule forecasting, and resource planning, lacking an integrated perspective [13]. These limitations contribute to persistent inefficiencies in coordination and communication among stakeholders, negatively impacting project performance and leading to significant material waste [14, 15]. Recent research has therefore explored the integration of emerging technologies, including IoT, artificial intelligence, and advanced analytics, to enhance data interoperability, real-time monitoring, and decision support in construction projects [16].

2.4 Business Intelligence Solutions for Construction Projects

Based on a market-oriented analysis of widely used ERP tools, Table 1 summarizes the main advantages and limitations of representative software solutions. The complexity and variability of construction projects limit the direct applicability of generic BI solutions, often requiring customization or sector-specific adaptations. Existing market solutions range from tools aimed at SMEs, offering basic project and financial management features, to more comprehensive platforms that integrate collaboration

Table 1 Comparison of ERP software solutions in the construction sector (author’s analysis)

Tool	Advantages	Limitations
Primavera Construction	Accounting, construction management, and HR integration; strong presence in Portugal; suitable for SMEs.	Relatively high licensing costs; learning curve for new users.
PHC CS Construção	Contract, works, and invoicing management; user-friendly interface; national solution.	Limited flexibility for external integration; moderate cost.
ETICADATA Obras	Well adapted to the Portuguese market; budgeting and construction monitoring modules.	Limited support for advanced analytics and BI integration.
SAP Business One (Construction)	Highly robust; integrates finance, logistics, and construction; scalable.	Very high costs; long implementation time; unsuitable for SMEs.
Microsoft Dynamics 365	Flexible and modular; native Power BI integration; strong analytics.	High licensing costs; requires construction-specific customization.
Autodesk Construction Cloud	Real-time collaboration; BIM support; document and schedule integration.	Focused on project management rather than financial control.
Procore	Strong project management focus; efficient team communication.	High cost; limited adaptation to the Portuguese context.
Buildertrend	SME-oriented; intuitive interface; cloud-based access.	Limited financial features; low penetration in Europe.

tools, Building Information Modeling (BIM) methodologies, and advanced analytics [9]. However, despite their technological maturity, these solutions frequently present drawbacks, such as high costs, limited flexibility, and insufficient alignment with the operational reality of small and medium-sized enterprises. Comparative analyses in the literature highlight these gaps and reinforce the need for modular, adaptable BI solutions tailored to the specific requirements of the construction sector.

3 Methodology

This section presents the methodology adopted throughout the project’s development, detailing the stages, techniques, and tools that supported the integration of the module into the existing Enterprise Resource Planning system. As this involves the design of a product, the methodological approach focuses on the processes of requirements analysis, modeling, architecture definition, development techniques, and solution validation procedures, based on established software engineering practices.

3.1 Requirements Analysis

In a meeting with the client, the functional requirements of the new modules were defined and the key performance indicators to support managers’ decision-making

were identified. The module should constitute an aggregator panel of key business information, allowing managers to obtain a global view in a single interface.

In general terms, indicators were established such as: active projects, completed projects, client with the highest number of projects, number of projects for that client, projects with the highest material costs, material classification, client classification, and supplier classification. These elements make up the Global Indicators tool. Within the specific scope of each project, two main indicators were defined: the time of extrapolation from the forecast and the comparison between the budget and the actual cost recorded.

To enable the calculation of the time extrapolation indicator, it was established that each project should be structured into chapters, each with its own budget and forecast and actual dates. This requirement led to the creation of two complementary tools: Chapter, in the Parameters group, responsible for defining the names and descriptions of the chapters; and Chapters per Project, in the Projects group, responsible for assigning chapters to projects, considering time and budget. Figure 1 presents the class diagram that illustrates the relationship between these concepts. The system as a whole has many other tables that store information about construction projects, equipment, employees, salaries, attendance, customers, suppliers, among others. However, the focus of this article is on performance indicators and not on explaining the end-to-end solution. Therefore, other tables are omitted.

3.2 Modeling

After the requirements analysis stage, interface mockups were created to validate the interaction flow and visual organization of the proposed solution. Prototyping tools such as Figma were used to build low-fidelity and medium-fidelity prototypes. Figure 2 shows the Budget Execution page template, where users can select a specific building, a date range and obtain detailed information about real costs in material, shared costs, direct labor, indirect labor, credit notes, and totals with and without taxes. In addition, there is space reserved for three graphs: the first is responsible for visually showing the proportion of construction costs over the months in bars, the second is responsible for comparing the execution period also in bars, and the third visually compares the budget with the amount executed.

Figure 3 shows the Global Indicators page template, where users can select a date range and obtain information on active projects during that period, the total number of projects completed by the end date, the clients with the highest number of projects, and the total number of projects associated with each client. Additionally, the interface displays classifications of materials, suppliers, customers, and projects based on actual cost criteria. There is a third model of specific indicators that allow the user to analyze time extrapolations in relation to forecast and actual start and end dates, as well as evaluate labor, materials and total costs, for a specific project in a specific chapter. However, this model was omitted due to article page limits.

The use of mockups allowed us to anticipate potential usability issues, adjust navigation elements, and ensure that the final solution was aligned with the client's expectations. In addition, these representations served as a direct reference for subsequent implementation steps.

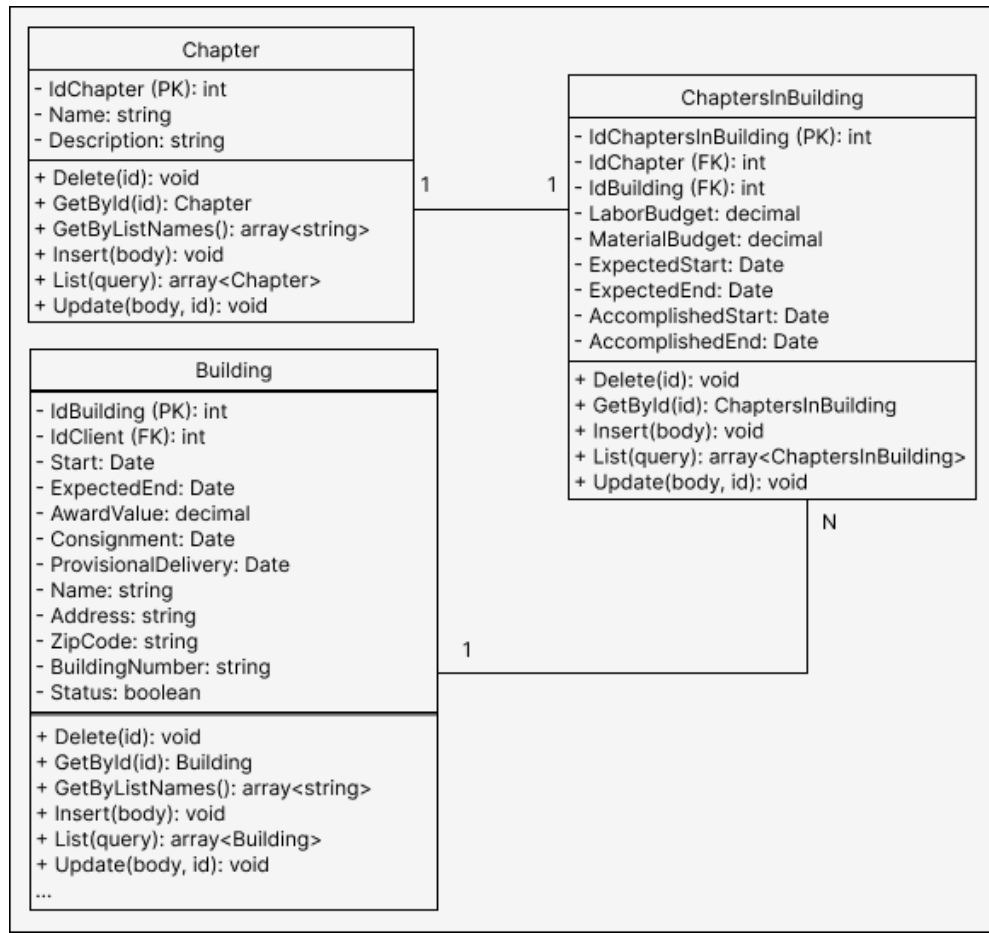


Fig. 1 Class Diagram for Chapters

3.3 Adopted Architecture and Agile Development Methodology

The system architecture is based, on the backend, on the C# language with the .NET framework, using Entity Framework Core to interact with the MySQL relational database. On the frontend, TypeScript, the React framework, and the Ant Design library were adopted. This technological combination made it possible to efficiently leverage the legacy structure of the original solution, developed in 2013.

The development methodology followed an agile approach, structured in Kanban and biweekly iterative cycles. This model enabled incremental management of activities, continuous prioritization of essential features, and rapid adaptation to new requirements [17]. This provided permanent visibility of the workflow, promoting



Fig. 2 Budget Execution Mockup

transparency, progress control, and early identification of bottlenecks. The organization of tasks into columns representing execution statuses allowed for systematic monitoring of the journey from design to delivery. The biweekly cycles reinforced regular planning, review, and adjustment moments, promoting continuous improvement, incremental validation of features, and constant alignment between technical objectives and project requirements.

3.4 Validation

The validation of the new modules was conducted continuously in collaboration with the client, through availability tests and weekly checks of the implemented features. Another important point is that there were expected values for part of the proposed calculations that were recovered manually before development. This also contributed to the verification of the final values obtained in the tables, cards, and graphs. More than just arriving at total values, it is essential that they are consistent with the records.

The interfaces and features developed were reviewed by comparing them with previously defined models and with the client's expectations regarding the modes of operation. This close collaboration throughout the execution process enabled the rapid identification of problems and the clarification of doubts associated with the business rules specific to the civil construction and administrative management sectors.

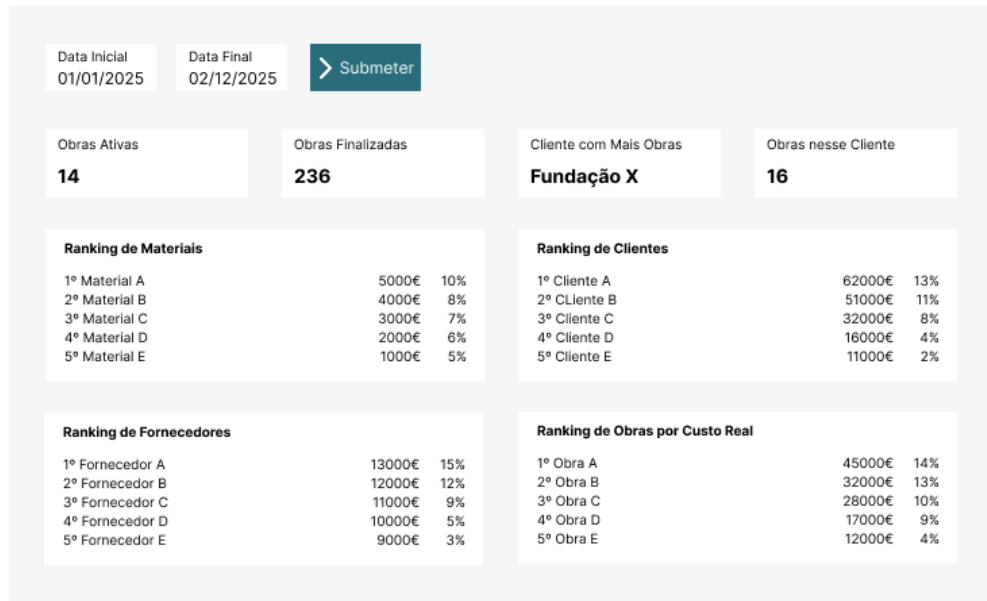
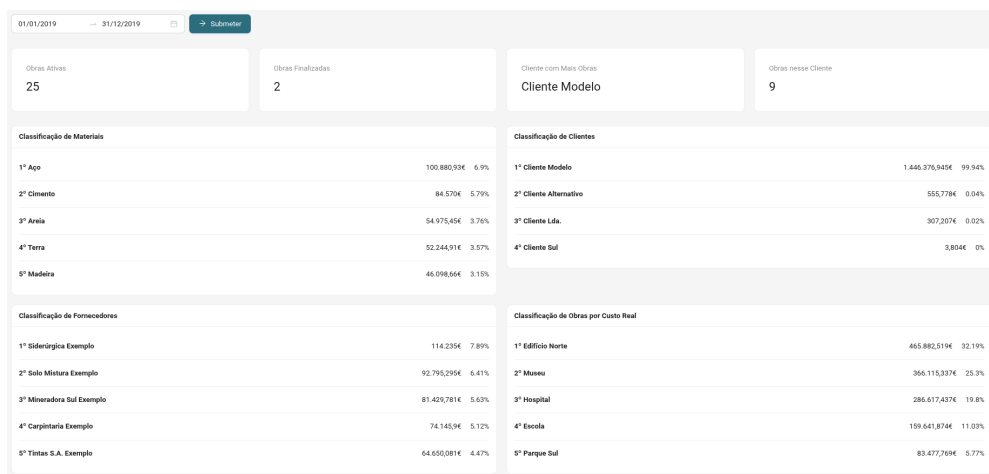
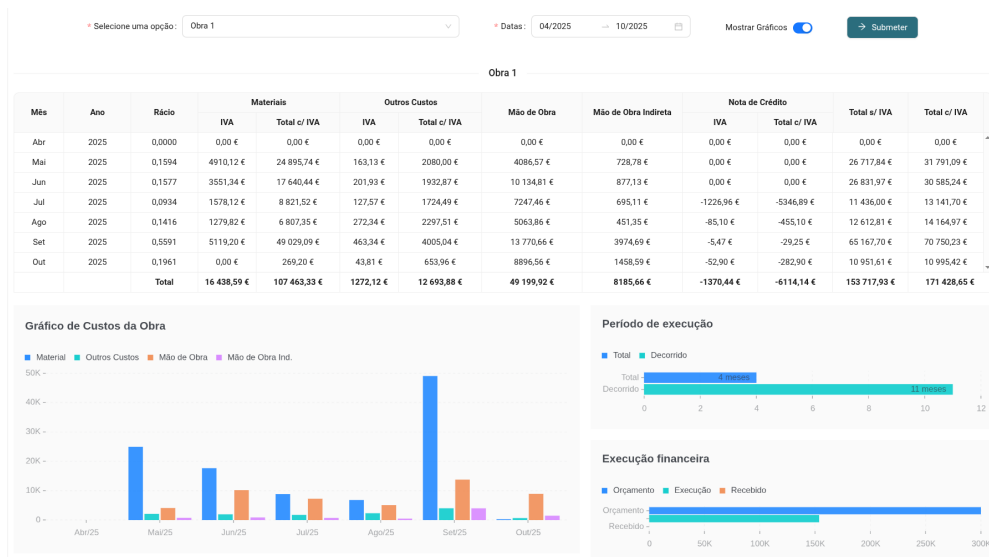


Fig. 3 Global Indicators Mockup

4 Results and Discussion

The dashboards include three main components: Budget Execution, Global Indicators, and Specific Indicators. On the Budget Execution page, Figure 4, the primary change was the introduction of an option to hide graphical elements. This allows users to focus exclusively on final values and improves performance when detailed visual analysis is not required. For the Global Indicators page, the implementation closely followed the mockup, Figure 5, with minor adjustments to visual scaling in order to better optimize the available screen space.



The Specific Indicators page, shown in Figure 6, includes the most visible visual changes. Colored borders were added to clearly distinguish between budget, execution, and comparison cards, improving interpretability and reducing cognitive load.

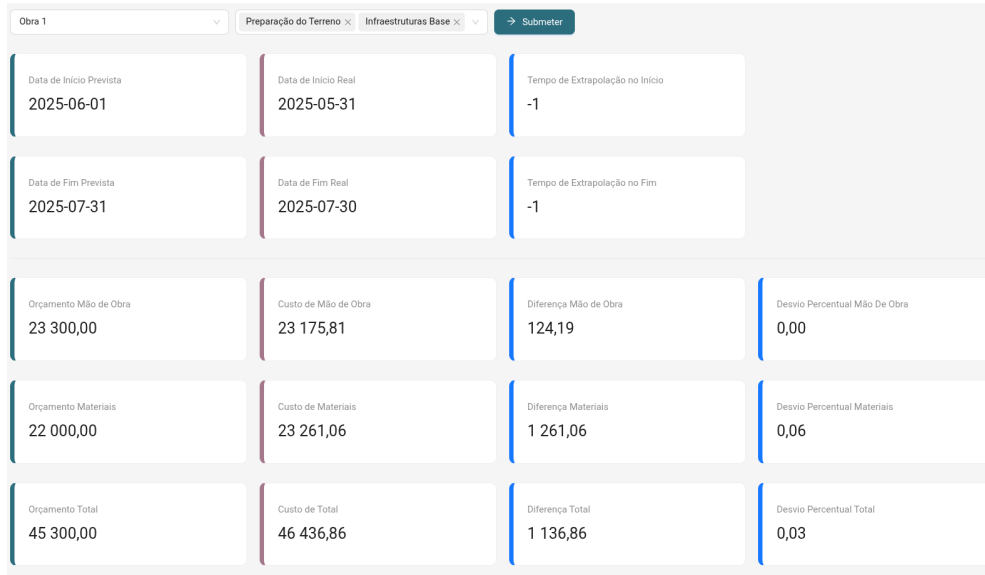


Fig. 6 Specific Indicators Page

without disrupting established workflows. In particular, the defined performance indicators and interactive dashboards effectively support budget execution monitoring, global performance assessment, and project-specific analysis, improving transparency and enabling earlier identification of cost and schedule deviations.

This study has some limitations. Validation was performed with a limited number of real-world tests at the client company, formal user satisfaction questionnaires were not completed, and the implemented dashboards cover a subset of the available operational data. These constraints reflect the current stage of development and the need to ensure high reliability for the initial production deployment. Despite these limitations, the results provide meaningful insights into the effectiveness of the BI module and its contribution to decision support for construction SMEs.

5 Conclusion and Future Work

The developed BI dashboards successfully transform raw operational data into actionable insights, advancing cost control, financial analysis, and project monitoring. They enable the identification of key materials, suppliers, customers, and projects, improve schedule tracking, and support comparisons between planned and actual budgets. By integrating organizational data into a clear, guided view, the BI module strengthens analytical capabilities, facilitates early detection of inefficiencies, and supports the digital transformation of construction SMEs through more timely and evidence-based decision-making. Future work will focus on incorporating an external module for automatically measuring attendance through online biometric registration.

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